

Experiment No: 04

Experiment Name: Design and Implementation of a 3-bit Synchronous & Asynchronous UP/DOWN Counter using Flip-Flops.

Objectives:

1. To understand the basic working principles of synchronous and asynchronous counters and the fundamental differences between them.
2. To design and implement a 3-bit synchronous Up, Down, and Up-Down counter using T flip-flops in MATLAB/Simulink or on a digital breadboard.
3. To design and implement a 3-bit asynchronous (ripple) Up, Down, and Up-Down counter using JK flip-flops.
4. To analyze and compare the propagation delay and timing behavior of synchronous versus asynchronous counters.
5. To construct the state transition table and timing diagrams for each counter type and verify the counter sequence experimentally.
6. To evaluate the effect of clock signal on the triggering mechanism of each flip-flop in both counter configurations.
7. To understand the role of T flip-flops and JK flip-flops in building binary counter circuits and apply Boolean logic in counter design.

Theory:

A counter is one of the most fundamental and widely used building blocks in digital electronics. In digital systems, a counter is a sequential circuit that goes through a predefined sequence of states in response to an input clock signal. Counters are used in a vast array of applications including frequency division, event counting, digital clocks, timers, memory address generation, and control sequencing in microprocessors and microcontrollers.

The basic building block of any counter is a flip-flop a bistable multivibrator that can store one bit of information and change its state in response to a triggering edge (rising or falling) of a clock signal. The most commonly used flip-flops in counter design are the JK flip-flop and the T

(Toggle) flip-flop. A T flip-flop toggles its output on every active clock edge when $T = 1$, while a JK flip-flop offers SET, RESET, HOLD, and TOGGLE operations based on its J and K inputs.

A 3-bit binary counter uses three flip-flops connected together and counts from 000 to 111 in binary (0 to 7 in decimal) for an up counter, or from 111 to 000 for a down counter. An up-down counter can count in both directions depending on a control input. The sequence of states each flip-flop output takes constitutes the state transition table of the counter, which is essential for analyzing its behavior.

There are several types of counters used in digital design: asynchronous (ripple) counters, synchronous counters, decade counters, ring counters, Johnson counters, up/down counters, and modulus counters. Each type differs in the way clock pulses are applied, the speed of operation, and the complexity of design. This experiment focuses on synchronous and asynchronous 3-bit binary up/down counters.

Asynchronous (Ripple) Counter:

An asynchronous counter, also called a ripple counter, is a type of digital counter in which the flip-flops do not all share a common clock signal. Instead, only the first flip-flop (least significant bit) receives the external clock pulse. Each subsequent flip-flop in the chain uses the output Q of the previous flip-flop as its clock input. As a result, the state change ripples through from one flip-flop to the next, one at a time hence the name “ripple counter.” Asynchronous counters are typically built using trailing-edge (negative-edge) triggered JK flip-flops, with $J = K = 1$ so that each flip-flop toggles on every falling edge of its clock input. The first flip-flop toggles every clock cycle; the second toggles every two clock cycles; the third toggles every four clock cycles. This creates the standard binary counting sequence. The key advantage of the asynchronous counter is its simplicity fewer connections and lower hardware complexity. However, because each flip-flop's output triggers the next, there is a cumulative propagation delay. In high-speed systems, this delay can cause glitches and incorrect intermediate states called “spurious states.” For a 3-bit counter, if the propagation delay per flip-flop is t_{pd} , the total worst-case delay is $3 \times t_{pd}$. This limits the maximum operating frequency of ripple counters and makes them unsuitable for high-speed or timing-critical applications.

3-bit Asynchronous Up Counter: In a 3-bit asynchronous up counter, the counter starts at 000 and increments by 1 with each clock pulse, reaching a maximum of 111 (decimal 7) and then

rolling back to 000. Each flip-flop toggles when the output of the preceding flip-flop transitions from HIGH to LOW (falling edge). Q0 is the LSB and Q2 is the MSB.

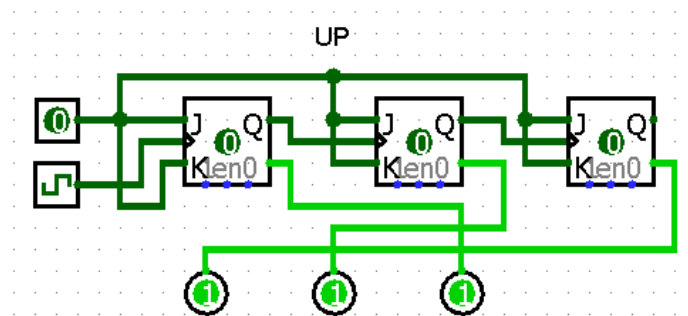


Figure 1: 3-bit Asynchronous Up Counter

3-bit Asynchronous Down Counter: A 3-bit asynchronous down counter counts from 111 (decimal 7) down to 000 and then repeats. In this configuration, each flip-flop is clocked by the complement output $\bar{Q}$ of the previous flip-flop rather than Q. This means the next flip-flop toggles when the previous output transitions from LOW to HIGH, producing a decrementing binary sequence.

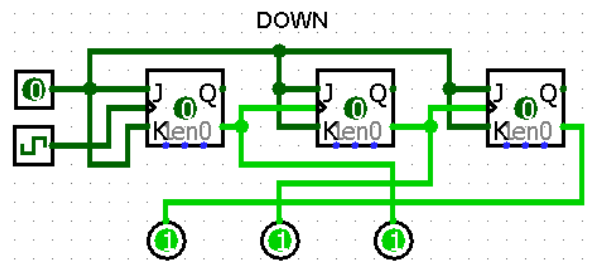


Figure 2: 3-bit Asynchronous Down Counter

3-bit Asynchronous Up-Down Counter: An asynchronous up-down counter combines both counting directions using a mode control input (M). When $M = 0$, the counter counts up by connecting Q outputs to subsequent flip-flop clocks. When $M = 1$, the counter counts down by connecting $\bar{Q}$ outputs instead. Logic gates are used to select the appropriate clock source for each flip-flop based on the mode input.

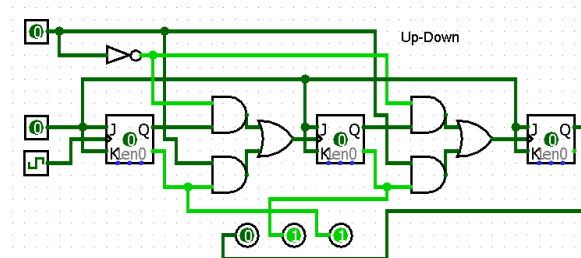


Figure 3: 3-bit Asynchronous Up-Down Counter

Synchronous Counter:

A synchronous counter is a type of digital counter in which all flip-flops share a single common clock signal. Every flip-flop receives the same clock pulse simultaneously, so their state changes occur at the same instant. This eliminates the propagation delay problem found in asynchronous (ripple) counters and allows synchronous counters to operate at much higher clock frequencies. In a synchronous binary counter, the least significant flip-flop (Q0) always toggles with every clock pulse. A flip-flop at any higher bit position toggles only when all lower-order bit positions are simultaneously HIGH (logic 1). This toggling condition is implemented using AND gates whose inputs are the outputs of all lower-order flip-flops. Synchronous counters are typically designed using T flip-flops due to the direct relationship between the toggle condition and the carry logic. The design methodology for a synchronous counter involves creating a state transition table listing all states and next states, deriving the excitation equations for each flip-flop, and minimizing those equations using Boolean algebra or Karnaugh maps. The result is a reliable, glitch-free counting circuit. Because all flip-flops change simultaneously, there are no spurious intermediate states, making synchronous counters the preferred choice in most modern digital system designs.

3-bit Synchronous Up Counter: In a 3-bit synchronous up counter, Q0 always toggles. Q1 toggles when Q0 = 1. Q2 toggles when Q0 AND Q1 are both 1. This creates the counting sequence 000 → 001 → 010 → 011 → 100 → 101 → 110 → 111 → 000. AND gates provide the carry logic for Q1 and Q2.

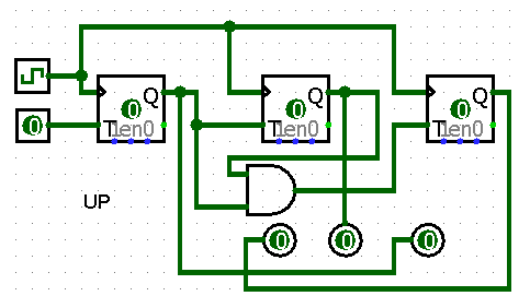


Figure 4: 3-bit Synchronous Up Counter

3-bit Synchronous Down Counter: In a 3-bit synchronous down counter, Q0 toggles every clock cycle. Q1 toggles when Q0 = 0 (i.e., $\bar{Q}0 = 1$). Q2 toggles when both Q0 and Q1 are 0. AND gates on the complemented outputs provide the borrow logic. The counting sequence is 111 → 110 → 101 → 100 → 011 → 010 → 001 → 000 → 111.

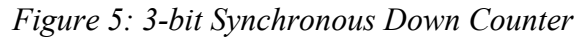


Figure 6: 3-bit Synchronous Up-Down Counter

Both synchronous and asynchronous counters play critical and complementary roles in digital system design. Understanding when and why to use each type is essential for any electronics or mechatronics engineer.

Importance of Synchronous Counters: Synchronous counters are the backbone of high-speed digital systems. Because all flip-flops are clocked simultaneously, there is no accumulated propagation delay, which allows synchronous counters to operate at significantly higher frequencies than asynchronous ones. This makes them indispensable in modern processors, FPGAs, digital signal processors, and high-speed communication systems. Synchronous counters

also avoid the spurious intermediate states (glitches) that occur in ripple counters, which is crucial in systems where incorrect transient states could trigger unintended operations.

Comparison Summary: In summary, asynchronous counters are valued for their simplicity and low-cost implementation in slow, non-critical applications, while synchronous counters are essential for fast, reliable, and precise digital operations. The choice between the two depends on the speed requirement, acceptable delay, hardware complexity, and the timing sensitivity of the target application. A thorough understanding of both counter types gives a digital designer the flexibility to make optimal design decisions in real-world engineering scenarios.

Discussion:

This experiment involved designing 3-bit synchronous and asynchronous up/down counters using flip-flops. The asynchronous counter, built with JK flip-flops, showed ripple delays due to sequential clocking. Each flip-flop received its clock from the previous stage's output, which introduced a cumulative propagation delay across the three stages.

Conclusion:

The experiment successfully demonstrated the design and operation of both 3-bit synchronous and asynchronous up/down counters. Synchronous counters implemented with T flip-flops provided accurate, glitch-free, and stable outputs by sharing a common clock, making them more reliable for high-speed applications. Asynchronous counters implemented with JK flip-flops, while simpler in design, exhibited the characteristic ripple delay that limits their use in timing-critical systems. Overall, the experiment improved understanding of digital counter design, ff behavior, the comparative timing characteristics of synchronous versus asynchronous sequential circuits.

References:

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